
Causal Steering of Protein Language Models: When Correlational Validation Fails, and How to Catch It

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Abstract

1 Activation steering has recently been shown to reproduce a known protein property
2 (thermostability) in ESM2-650M via difference-of-means vectors. We ask whether
3 a scoring proxy’s correlation with real experimental labels predicts whether steering
4 toward it actually works, and find the answer is no. Under a pre-registered
5 six-criterion protocol testing five novel target properties, our best-validated proxy
6 (binding affinity, $r = 0.80$ held-out) steers nothing, while a substantially weaker
7 one ($r = 0.22$, catalytic activity) produces a robust, three-seed-replicated effect.
8 This mirrors an independent, component-level finding in the same model family:
9 attention heads ranked by correlation with real 3D contacts show near-zero re-
10 lationship to their causal ablation effect. We show a single-seed steering result
11 can mislead in criterion-specific ways – one target’s effect direction replicates
12 across three seeds while its residue-exclusion check does not (2 of 3) – and in-
13 troduce a lightweight diagnostic, steering-vector cosine similarity across targets,
14 that explains an otherwise-unexplained ambiguous result as a geometric echo of
15 a different property’s validated direction. We release the full protocol, all target
16 harnesses, and the seed-replication data.

17 1 Introduction

18 Difference-of-means activation steering – adding a direction built from the mean activation difference
19 between high- and low-property example groups to a residual stream – reproduces known thermosta-
20 bility biology in ESM2-650M [Huang et al., 2025]. We ask the natural follow-up: does a scoring
21 proxy’s correlation with real labels predict whether steering toward a *new* property works? Across
22 five novel targets under one pre-registered protocol, the best-validated proxy in our study ($r = 0.80$,
23 binding affinity) produces a flat null effect, while a substantially weaker proxy ($r = 0.22$, catalytic
24 activity) produces a clean, three-seed-replicated effect. This mirrors an independent, component-level
25 result: Vig et al. [Vig et al., 2021] found an attention head whose correlation with real 3D contacts is
26 $12.9\times$ enrichment, explicitly flagged as untested causally; ablating it gives an effect indistinguishable
27 from a low-correlation control, and a full 480-head sweep ranks the correlational top pick 313th of
28 480 by actual causal effect. Two levels, two techniques, one lesson: correlational signal is evidence
29 about representations, not about intervention. Our contribution is the audit discipline needed to trust
30 a causal claim here, demonstrated via two near-misses caught in our own pipeline before they could
31 be reported as clean results.

Table 1: Five novel targets. Proxy r : Pearson correlation vs. real labels, held-out, before any steering run.

Target	Proxy r	Steering result	Verdict
Binding affinity	0.80	flat null, all α	KILL
Catalytic activity	0.22	PASS, 3/3 seeds	PASS
Intrinsic disorder	0.45	3/3 direction; 2/3 robustness	PASS*
Immunogenicity	≤ 0.10	not run	KILL (pre-run)
Expression yield	0.31	fails residue-exclusion	AMBIGUOUS

*PASS on criteria 1,2,4,6, and 2/3 seeds on criterion 3.

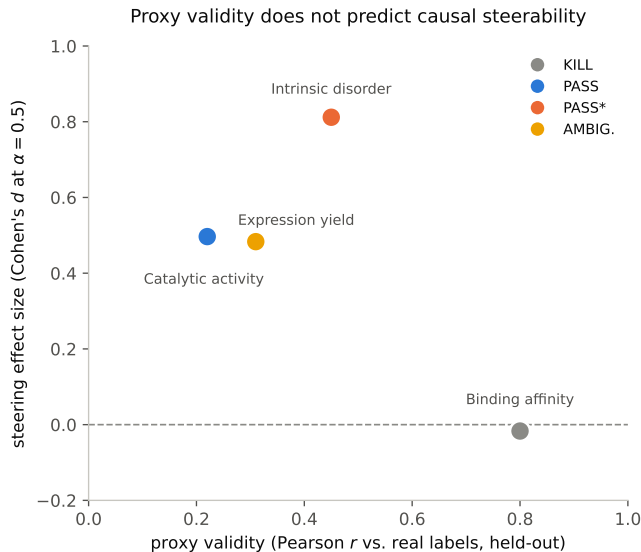


Figure 1: Proxy validity vs. steering effect size (Cohen’s d , $\alpha = 0.5$). The best-validated proxy (binding affinity, $r = 0.80$) gives the smallest effect; a proxy less than half as strong (catalytic activity, $r = 0.22$) gives one of the two largest. No positive trend.

32 2 Method

33 We lock six criteria before testing any target, computed automatically: (1) the real direction beats a
34 matched-norm random-direction control via paired bootstrap (10,000 resamples); (2) a monotonic
35 dose-response across ≥ 3 steering strengths, restricted to a pre-established safe range (§4); (3) the
36 effect survives residue-exclusion; (4) the proxy is validated against real labels *before* the run; (5) it
37 beats the best existing technique where one exists; (6) ≥ 150 held-out evaluation sequences. A target
38 failing any criterion outright is a **KILL**; one passing 1–2 but failing 3 or 4 is **AMBIGUOUS**.

39 3 The Central Finding: Proxy Validity Does Not Predict Causal Effect

40 **Binding affinity** (ProteinGym DMS, KRAS/DARPin): a mutational-sensitivity-weighted proxy
41 validated more rigorously than any other target here ($r = 0.80$ held-out; weight-shuffle null $r =$
42 -0.001 ; held-out-position generalization $r = 0.54$; single-to-double-mutant extrapolation $r = 0.70$)
43 – yet the steering effect is a flat null at every α , zero degenerate sequences at any strength. A plausible
44 mechanism: this dataset’s vector-building groups are single-backbone point mutants (1–2 residues
45 from a shared reference), giving difference-of-means little compositional signal to act on – measured
46 directly, mean compositional distance between low/high groups is 0.003 here vs. 0.02–0.05 for the
47 three cross-protein targets below.

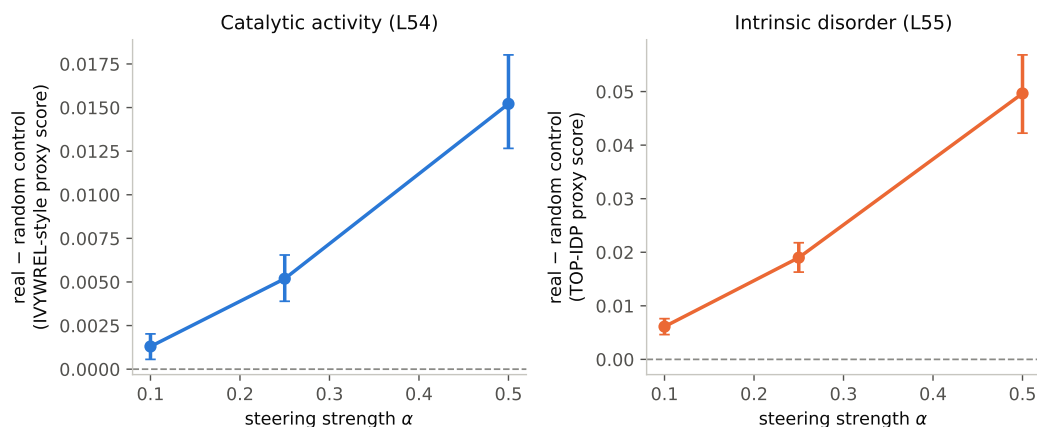


Figure 2: Dose-response, the two targets passing criterion 2. Both monotonic, 95% bootstrap CIs bounded away from zero throughout.

48 **Catalytic activity** (DLKcat, cross-protein kcat): a glycine-minus-arginine proxy, the weakest of the
 49 four runnable targets ($r = 0.22$), matching the published activity-stability tradeoff in enzymology.
 50 Produces a clean, monotonic, residue-robust effect (Figure 2, left) replicated identically across three
 51 independent seeds – the one target with fully unanimous seed-robustness.

52 **Intrinsic disorder** (DisProt, TOP-IDP scale, $r = 0.45$, the strongest proxy here, AUC 0.71 at the
 53 per-residue level): the effect is significant and monotonic on all 3 seeds (Figure 2, right), but its
 54 residue-exclusion check passes on 2/3 seeds (36–42% magnitude retained) and fails on the third
 55 (10% retained, CI crosses zero) – existence and direction are seed-stable; the artifact check is not,
 56 uniformly.

57 **Immunogenicity** (IEDB, 4 tiers): proxies predictive on a surrogate binding-affinity tier ($r = 0.37$ –
 58 0.43) collapse near-chance on the real T-cell-response endpoint ($r \leq 0.10$); a full-length-antigen
 59 proxy reaching $r = 0.38$ is a source-organism confound (organism identity explains 58% of the
 60 apparent signal; flips to $r = -0.32$ once held out of cross-validation). Killed before any steering run.

61 **Expression yield** (eSol, net-charge-deviation proxy, $r = 0.31$ – 0.34): significant, dose-responsive,
 62 but collapses under residue-exclusion. Rather than discarding this, we compute steering-vector
 63 cosine similarity against every other target: $+0.38$ overall with intrinsic disorder’s vector, rising to
 64 $+0.56$ – 0.67 at the three deepest layers – consistent with disordered regions independently reducing
 65 solubility/expression in the literature. This is best read as a partial echo of disorder’s already-validated
 66 effect through a more collapse-prone proxy, not a sixth steerable property.

67 4 Self-Correction Case Studies

68 **A spurious pass from an unsafe comparison.** Testing whether restricting steering to the 5 layers
 69 previously found causally necessary preserves the full 33-layer effect, a first pass selected its best-
 70 performing α from the whole tested range, including $\alpha \geq 1.0$, and reported a clean pass at $\alpha = 2.0$.
 71 This was comparison artifact: at that α the 33-layer arm had already fully collapsed into degenerate
 72 output (zero non-degenerate sequences), while the 5-layer arm had not – it won only by breaking
 73 down later. Restricting the comparison to a range where neither arm has collapsed and rerunning end
 74 to end: the 5-layer subset is real but retains only 43–59% of the full effect.

75 **A criterion passing on only 2 of 3 seeds.** The disorder target’s residue-exclusion check (§3) is
 76 seed-sensitive even though the headline effect is not – a single seed could have reported either a false
 77 unanimous pass or a false kill on this one criterion, undetectable without a multi-seed rerun.

78 5 Related Work

79 Huang et al. [2025] validate this steering method on thermostability; we extend it to five new
80 properties under a pre-registered protocol and test, to our knowledge for the first time, whether proxy
81 correlation strength predicts steering success across unrelated properties. Vig et al. [2021] identify
82 the head we redo as a genuine causal test. Adams et al. [2025] train SAEs on ESM-2 and show via
83 linear probing – correlational – that thermostability and localization determinants are identifiable in
84 SAE feature space; their one steering check is a coherence test, not validation against a real label, and
85 their discussion states “investigating how to steer the model in useful ways remains an open direction
86 for future work.” We answer a version of that for a non-SAE mechanism, specifically for whether
87 correlational validity predicts causal usefulness.

88 6 Discussion

89 Every steering result here is causal about the model’s activation space – a real intervention against
90 a random-direction control and a residue-exclusion check – not validated against a wet-lab assay:
91 a PASS means “causally steers a validated proxy,” not “causally engineers the biological property.”
92 The vector-geometry check that explained the expression-yield result is cheap (one extra embedding
93 pass) and worth running by default whenever multiple targets are tested together. All experiments use
94 one model (ESM2-650M) at one scale; all proxies are compositional heuristics; each target’s proxy
95 was selected as the best of several candidates on the same held-out split its reported r comes from,
96 inflating every reported r (values should be read as upper bounds); we ran many significance tests
97 without a multiple-comparisons correction, partly mitigated by requiring multi-alpha/seed agreement
98 for a PASS; seed-robustness was checked for two of five targets; no structural/functional plausibility
99 check was run on generated sequences.

100 7 Conclusion

101 Correlational validity – of an attention head’s biological alignment, or a proxy’s correlation with
102 real labels – does not predict causal effect when actually tested, at two levels of the same model
103 class. Proxy validation strength is not a reliable filter for which targets are worth attempting, and a
104 single steering run should not be trusted without a seed-robustness check where the result matters
105 downstream.

106 Responsible-use statement

107 This work identifies methodological pitfalls in causally validating activation-steering claims on protein
108 language models and does not introduce a new capability for generating harmful biological sequences;
109 all target properties studied (thermostability, binding, catalysis, disorder, immunogenicity, expression)
110 are common protein-engineering objectives with dual-use risk no greater than the underlying published
111 steering method [Huang et al., 2025] we extend. We see no additional mitigation specific to this
112 paper’s contribution beyond standard biosecurity practice already governing protein-language-model
113 research.

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